

3.6.5 Riekkola, Lewis, and Porter Point Units

Riekkola, Lewis, and Porter Point Units consist primarily of Ocosta silty clay loam in the diked pastures and marsh areas. Forested areas in the higher elevations surrounding the units, including Lewis Hill, consist of Palix silt loam and some Ilwaco silt loam.

3.6.6 Leadbetter Point Unit

Leadbetter Unit has five soil types. The outer beach above mean high tide is classified as beach, this area has no vegetation and is subject to continual wave action during high tide. Dune land makes up the majority of outer peninsula westward and north. The dune land is very deep fine sand, drains excessively, and is constantly shifted by strong coastal winds. The dune land topography consists of a primary foredune, an interdune area of dunes and hollows and foredune ridges that run parallel with prevailing winds from the shoreline (also called the foredune complex), and a relatively flat deflation plain still further inland. The water table is at the surface of the interdune area during the winter months. The beachgrass and lodgepole pine areas of the peninsula tip and interior of Grassy Island is Westport fine sand, which forms on slightly weathered stabilized sand dunes. Westport fine sand is covered in a thin mat of organic material and is also very deep and excessively drained. The protective organic layer of Westport fine sand is extremely fragile. The dune land and Westport fine sand are highly susceptible to wind erosion when exposed.

The salt marsh and most of the southern portion of Grassy Island are Ocosta silty clay loam. The forested area near the Leadbetter parking area is Yaquina loamy fine sand, a very deep, somewhat poorly drained soil. The water table in winter is from 24 inches deep to the surface of this soil. The duff layer is half an inch thick (USFWS 2003b).

3.6.7 Wheaton Unit

Wheaton Unit has three soil types: Arta silt loam, Grehelam silt loam, and Rennie silty clay loam. Arta silt loam is a very deep, moderately well-drained soil present in the eastern hay field and higher areas of the homestead site and field. Runoff is slow and the hazard of water erosion is slight in Arta soil. Arta soil supports hemlock and red alder forest but is presently maintained as pasture. Grehelam silt loam is also a very deep, well-drained soil found on the natural dikes of the floodplain, which makes up the majority of the unit and includes the west field that is across the Willapa River and the field that is between Mill Creek and the Willapa River. Grehelam soil is subject to brief periods of flooding in winter. Grehelam soil typically supports Douglas fir and red alder forest but is maintained as pasture. A small area of Rennie silty clay loam exists in the oxbow wetland on the north side of the Willapa River, near the bend in the boundary fence. Rennie soil is very deep, poorly drained soil occurring on the floodplain. Permeability is slow, the water table is high seasonally and runoff is very slow in this soil, resulting in the formation of small seasonal wetlands.

quality rating, with higher quality wetlands requiring greater buffer protection zones. The Category I wetlands will have a designated 100-foot buffer zone, whereas a Category IV wetlands will have a 25-foot buffer protection zone. The wetlands found on the site will also be subject to Federal and state removal/fill wetland regulations if impacted.

4.3.9 Coastal Dunes and Beaches

Sand beaches with associated dunes dominate the southern Washington Pacific coastline, while the northern coast is more rugged and rocky with steep headlands and numerous offshore islands and rocks. Willapa Bay and Grays Harbor are two large bays located along the outer coast in the southern half of the state. Historically low hummock sand dune formations characterized by large areas of open sand, formed by sparsely vegetated native dune plant species. Coastal marine and wind processes worked to maintain native plant communities in early successional stages on the outer prism of many of these beaches. Where dunes were more stable and blowouts less frequent, a mosaic of native prairie and dune grasslands, freshwater lakes, swamps, bogs, and spruce-dominated forests developed. High rainfall maintained high water tables favorable for plant growth.

Invasive, non-native beach grasses (*Ammophila breviligulata* and *A. arenaria*) planted to stabilize dune community have change dune morphology and native plant communities. Mild climate allows vegetation to establish easily and rapidly. Accelerated succession due to fire suppression progresses from herbaceous beachgrass, to shrub (often invasive non-natives such as Scotch broom and common gorse), to pioneer lodgepole pine or climax Sitka spruce forest. The Columbia River once created extensive sediment transport and ocean currents influenced by a log-shore drift deposited sediment continually nourishing the coastal sand beaches. Dams on the Columbia River have altered sediment loads, and jetties at the river mouth and entrances to the bays have altered sediment transport.

The endangered pink sandverbena (*Abronia umbellata*) and other rare native dune plants like yellow sandverbena (*Abronia latifolia*), gray beach pea (*Lathyrus littoralis*) and beach morning glory (*Convolvulus soldanella*) are found along the sparsely vegetated sand beaches and coastal dunes within the Refuge where the spread of non-native beachgrass is controlled or kept in low densities due to the influence of naturally occurring erosion processes.

The Leadbetter Point Unit lies at the northern tip of the Long Beach Peninsula, at the mouth of Willapa Bay, in Pacific County, Washington. The Long Beach Peninsula separates the Pacific Ocean from Willapa Bay. The west side of the area is characterized by open windswept beaches backed by low vegetated dunes. The tip of the peninsula was largely barren sand, and the east side consists of a narrow beach with a few small, sheltered openings cut into the beachgrass by high water in winter. A small, isolated portion of beach exists to the east, on Willapa Bay, and is referred to as Grassy Island although it is attached to the peninsula.

The northern end of the Long Beach Peninsula was in a state of gradual northward accretion from at least 1965 to 1999. Invasion of American beachgrass and European beachgrass has followed accretion, progressively filling in the dunes behind the sand spit. In conjunction with slowed accretion in more recent years, the vegetation line has moved westward and the vegetation-to-water distance has decreased (Phipps 1990) resulting in a narrower beach and

probably less suitable plover habitat. Recent maps from the Washington State Department of Transportation show that the tip of Leadbetter Point has been gradually eroding since mapping efforts began in 1999. As the tip has eroded, the peninsula to the southwest has gotten wider. Leadbetter Point is one of the northern-most breeding sites for the western snowy plover (*Charadrius alexandrinus nivosus*) on the Pacific Coast (Jaques 2001).

The unique natural elements protected at Leadbetter Point include salt marsh, native dunegrass, lodgepole pine (shore pine) forest, shrub/lodgepole pine (shore pine), and open beach habitats. Leadbetter Point contains high-quality examples of high-salinity Virginia glasswort/inland saltgrass marsh, low salinity marsh, and transition zone wetlands. Flora associated with the marshes are of primary significance, as are the dune grassland and deflation plain communities. Pockets of native plants within the secondary dune, deflation plains, and dune troughs are also significant ecological features and are of high quality compared to these remaining plant communities in Washington. The open beach and dune grassland communities of Leadbetter Point have been significantly impacted by the invasion and naturalization of two non-native beach grasses. The salt marsh has been invaded by smooth cordgrass, an eastern salt marsh species, although efforts to control cordgrass in recent years have essentially eliminated it from Leadbetter Point. Selective removal or control of plant species not native to Leadbetter Point, including *Spartina*, Scotch broom, and common gorse, was an approved management activity at the time the RNA was established. Removal and control of the non-native beach grasses has been approved and work has been done as part of the management of habitat for the federally threatened/state endangered western snowy plover (Caicco 1989; Willapa NWR files).

4.3.10 Grasslands and Short-grass Fields

Native grasslands occurred historically on the Long Beach Peninsula. Currently there are very few of these native plant communities remaining. Where grasslands still exist they are often pastures of introduced grasses, and sedges in wetter areas, managed as livestock rangeland, golf courses and residential lawns. Willapa NWR is planning to develop a habitat restoration project to create early successional, coastally influenced grassland habitat for the Oregon silverspot butterfly. WDFW has already implemented a similar project at two small sites on state land on the Long Beach Peninsula.

Proximity to the salt spray from the ocean, mild temperatures, high rainfall, and fog have maintained the low-growing, open natural grasslands by suppressing encroaching trees and shrubs. Natural processes responsible for sustaining the community structure and composition are wind transport of sand, small mammal activity, herbivory, and fire.

Habitat loss has resulted from dune stabilization caused by the introduction and spread of non-native beachgrass that encourages rapid succession to forested habitats. The early blue violet (*Viola adunca*), a host plant of the Oregon silverspot butterfly larvae, and other native grasses and forbs, are out-competed by the introduced grasses and herbs and shaded out by weedy shrubs and expanding pioneer lodgepole pine forests. As coastal areas become stabilized and developed, the influence of natural processes that sustain native habitats is reduced or eliminated.

The Refuge currently has several managed short-grass pastures in the South Bay Units totaling 250.5 acres.

About the Leadbetter Point Unit

Historically, habitats along the Long Beach Peninsula consisted of low hummocky sand dune formations characterized by large areas of open sand with sparsely vegetated native dune plant species. Coastal marine and wind processes worked to maintain native plant communities in early successional stages on the outer prism of the beach. The dunes were more stable and blowouts less frequent; a mosaic of native prairie and dune grasslands, freshwater lakes, swamps, bogs, and spruce-dominated forests developed. High rainfall maintained high water tables favorable for plant growth.

The Leadbetter Point Unit encompasses 1,742 acres and is located at the northern tip of the Long Beach Peninsula, near the mouth of Willapa Bay. The coastal dune habitats consist of sand dunes in various stages of ecological succession including bare unstable sand; beachgrass-covered dunes; a transition zone composed of shrubs, small lodgepole pine (shore pine), and grass; lodgepole pine (shore pine); freshwater wetlands; and salt marsh.

At one time, the Columbia River provided the coastal shoreline with an extensive transport load of sediment; the ocean currents influenced by a long-shore drift deposited the sediment creating and maintaining the coastal sand beaches. Today, dams on Columbia River have altered sediment loads, and jetties at the river mouth and entrances to the bays have altered sediment transport along the coast. The beaches no longer have the natural processes depositing large amounts of sand and sediment necessary to maintain the sand beaches and dune habitats for a variety of native plants.

The habitat has changed in recent history with an accelerated plant succession that is also due to fire suppression efforts. These efforts have encouraged a plant successional progress away from the historic herbaceous beach grass, to a shrub (often invasive non-natives such as Scotch broom and common gorse) habitat leading to a pioneer lodgepole pine or climax Sitka spruce forest.

The west side of the unit is characterized by open wind-swept beaches backed by vegetated dunes. The extreme tip of the peninsula is largely barren sand, and the east side consists of a narrow beach with a few small, sheltered openings cut into the beachgrass by high water in winter. A small, isolated portion of beach exists to the east, on Willapa Bay, and is referred to as Grassy Island although it is attached to the peninsula.

The northern end of the Long Beach Peninsula was in a state of gradual northward accretion from 1965 to 1999. Invasion by non-native beach grasses has followed accretion, progressively filling in the dunes. In conjunction with slowed accretion in more recent years, the vegetation line has moved westward and the vegetation-to-water distance has decreased resulting in a narrower beach. Recent maps from the Washington State Department of Transportation show that the tip or northern portion of the unit has been gradually eroding since mapping efforts began in 1999. As the tip has eroded, the peninsula to the southwest has become wider.

American dunegrass (*Leymus mollis* or *Elymus mollis*), a native dunegrass, exists in small patches on the Refuge unit. Two invasive non-native beachgrass species, American beachgrass and European beachgrass (*Ammophila breviligulata* and *A. arenaria*), planted to stabilize dunes, have changed historical dune morphology and native plant communities. American beachgrass is the most abundant of the three grass species on the Long Beach Peninsula dunes, although all

three species can be found growing together and there are patches of these species growing separately. The beachgrasses form a continuous band of vegetation parallel to the high tide mark along the outer ocean beach.

Non-native beachgrasses out-compete native vegetation, alter the dune ecosystem, and form dense stands that reduce the amount and quality of nesting habitat for native wildlife, including the federally threatened, state-endangered western snowy plover and a state-endangered, Federal candidate species, streaked horned lark. Non-native beachgrasses have rapidly taken over a majority of formerly open sand dunes that provide nesting habitat for these two species. Western snowy plover numbers have declined along the U.S. Pacific coast due to habitat degradation as well as impacts from the expanding predator populations. One of the most significant causes of habitat loss for coastal breeding population of western snowy plovers has been the encroachment of introduced beachgrasses.

The invasion of non-native beachgrasses has also caused a dramatic reduction of coastal native plants and is a primary threat to the state endangered pink sandverbena (*Abronia umbellata*) which is also a Federal species of concern. Pink sandverbena and other rare native dune plants like yellow sandverbena (*Abronia latifolia*), grey beach pea (*Lathyrus littoralis*) and beach morning glory (*Convolvulus soldanella*) are found along the sparsely vegetated sand beaches and coastal dunes.

In order to protect and encourage native plant growth the Refuge has implemented a Coastal Habitat Restoration Project. This ongoing project was initiated in 2002 and has continued each year with successful results. The mechanical and chemical removal and control of non-native beachgrass has resulted in over 120 acres of restored habitat that has successfully attracted nesting western snowy plovers and streaked horned larks. Oystershell was added to portions of the cleared area to provide camouflage for nests and reduce blowing sand to protect the bird nests. This habitat restoration area supports the only known population of pink sandverbena in Washington State; this plant species was thought to be extirpated in the state (Federal species of concern, Washington State endangered species). In 2006, it was able to re-establish itself, from a long-term seed bank, because beachgrass had been removed from the site. Thousands of plants now exist at the site due to transplantation of propagated individuals and broadcast seeding efforts as well as a high success rate due to natural seeding.

Leadbetter Point Research Natural Area (RNA) is located entirely in the Refuge and was put on the Washington Register of Natural Areas in 1989. The original designation included 1,705 acres of the peninsula tip, Grassy Island, and the marsh between the island and peninsula tip; however, the unit is now approximately 1,742 acres due to sand accretion at the peninsula tip. This area represents the highest quality habitat, largest coastal sand dune ecosystem in Washington State.

The natural elements protected include salt marsh, native dunegrass, lodgepole pine (shore pine) forest, shrub/lodgepole pine (shore pine), and open beach habitats. The bay side of the unit contains some of the most significant saltmarsh habitats remaining in Washington. It also contains high-quality examples of high salinity Virginia glasswort/inland saltgrass marsh, low salinity marsh, and transition zone wetlands. Flora associated with the marshes are of primary significance, as are the dune grassland and deflation plain habitat communities. Pockets of native plants within the secondary dune, deflation plains, and dune troughs are also significant